

New antiretroviral inhibitors and HIV-1 drug resistance: more focus on 90% HIV-1 isolates?

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Abstract

Combined HIV antiretroviral therapy (cART) has been effective except if drug resistance emerges. As cART has been rolled out in low-income countries, drug resistance has emerged at higher rates than observed in high income countries due to factors including initial use of these less tolerated cART regimens, intermittent disruptions in drug supply, and insufficient treatment monitoring. These socioeconomic factors impacting drug resistance are compounded by viral mechanistic differences by divergent HIV-1 non-B subtypes compared to HIV-1 subtype B that largely infects the high-income countries (just 10% of 37 million infected). This review compares the inhibition and resistance of diverse HIV-1 subtypes and strains to the various approved drugs as well as novel inhibitors in clinical trials. Initial sequence variations and differences in replicative fitness between HIV-1 subtypes pushes strains through different fitness landscapes to escape from drug selective pressure. The discussions here provide insight to patient care givers and policy makers on how best to use currently approved ART options and reduce the emergence of drug resistance in ~33 million individuals infected with HIV-1 subtype A, C, D, G, and recombinants forms. Unfortunately, over 98% of the literature on cART resistance relates to HIV-1 subtype B.

Keywords: antiretroviral therapy, low-and middle-income countries, HIV-1 genetic diversity, discovery of novel HIV inhibitors, HIV-1 subtype specific resistance, non-subtype B HIV-1

Introduction to antiretroviral treatment

The discovery that AIDS was caused by HIV-1 in 1983 led to the rapid testing of nucleoside analogs that had been shown to inhibit avian and murine retroviruses. 3'-azido-3'-deoxythymidine (AZT), first synthesized in 1964, was the leading agent among nucleoside reverse transcriptase inhibitors (NRTIs) and was rapidly approved (Fig. 1) for HIV-1 treatment in 1987 after undergoing a small exploratory study and a double-blind placebo-controlled trial (CDC 1981, Gottlieb et al. 1981, Masur et al. 1981). However, drug resistance quickly emerged in people living with HIV (PLWH) taking AZT (Larder, Darby and Richman 1989). In subsequent years, more NRTIs were available for HIV-1 treatment, including didanosine (2',3'-dideoxyinosine) and zalcitabine (2',3'-dideoxycytidine), which allowed the comparison of treatment with dual NRTIs to monotherapy (CDC 1981, Gottlieb et al. 1981, Masur et al. 1981). Although drug resistance still emerged, better outcomes in antiretroviral therapy-naïve PLWH were recorded with dual nucleoside analog therapy, paving the way for triple therapy—highly active antiretroviral therapy (HAART)—in 1996 when a new class of antiretroviral, protease inhibitors were approved (Darbyshire and Aboulker 1996, Hammer et al. 1996, Katlama 1996, Schooley et al. 1996). A number of studies, including ACTG 320, showed that combining the protease inhibitor indinavir with two NRTIs, one of which was lamivudine (3TC), provided prolonged control of low level viremia (Gulick et al. 2000). Currently, there are over 34 individual therapeutics and 23 drug combinations approved by the

FDA for HIV-1 treatment, as well as one cART combination for HIV-1 prevention (Fig. 1).

Since their first use in the mid 1990 s, cART has been improved and refined for better treatment outcomes, increased potency, fewer adverse events, higher barriers to drug resistance, and reducing pill burden from upwards of 9 pills/day to single daily pill formulations. These new formulations include fixed-dose combinations with 2 to 4 active antiretroviral drugs in single tablet (e.g. abacavir (ABC)+lamivudine (3TC)+dolutegravir (DTG); Fig. 1) and injectables with long acting formulations of specific antiretroviral drugs including recently approved cabotegravir/rilpivirine and cabotegravir for the treatment and prevention of HIV, respectively (GSK 2021; UNAIDS 2021). These long-acting formulations are dosed once per week or once per month, are well tolerated, and provide improve adherence to cART. Other long acting formulations and delivery systems are still in clinical trials and include dapivirine within a vaginal ring, and tenofovir alafenamide (TAF) as a subdermal implant (Gunawardana et al. 2015, Baeten et al. 2016, Margolis et al. 2017). The success of current antiretroviral drugs (ARVs) has also been used as HIV-1 prevention tool among the risk population. In the PARTNER1&2 trials, a near-complete block in transmissions (upper confidence limit 95%, 0.23–0.3 per 100 years of follow up) were reported between discordant couples if the infected partner received cART and had undetectable viral loads (Rodger et al. 2016, Rodger et al. 2019).

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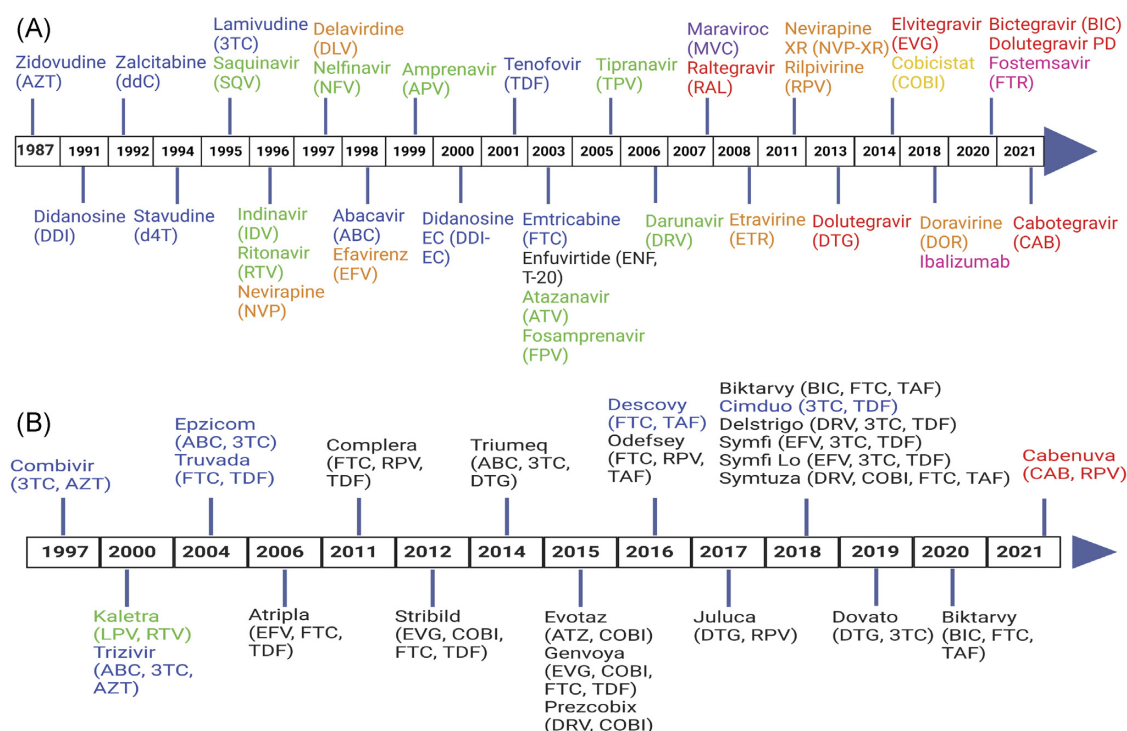


Figure 1. FDA-approved HIV-1 treatment drugs and combinations. **(A)** Regimens approved for HIV-1 treatment: blue, NRTIs; orange, NNRTIs; green, PIs; red, INSTIs; yellow, pharmacokinetic enhancers; pink, post-attachment inhibitors; and black, fusion inhibitors. **(B)** Brand names of approved oral and injectable ART combinations with respective drug abbreviations: blue, combinations consisting of reverse transcriptase inhibitors; green, combinations of PIs; red, combination of INSTIs and NNRTIs; black, combinations with more than one class of drug.

Accessing use of cART in the global pandemic: Factoring in socioeconomic difference and HIV-1 genetic diversity

HIV-1 is mainly divided into 4 groups: group M (major), O (Outlier), N (non-M, non-O) and P (Sharp and Hahn 2011). Group M is the most widely distributed and is further divided into 9 subtypes: A-D, F-H, J, and K (Sharp and Hahn 2011). Globally, subtype C accounts for the majority of HIV-1 infections (48%), although subtype A (12%), subtype B (11%), CRF02_AG (8%), CRF01_AE (5%), subtype G (5%), and subtype D (2%) are also prevalent. HIV-1 subtype C is commonly found in Southern Africa, Eastern Africa, Brazil, China, and India, with rapidly increasing prevalence seen in regions of Southwest China, Malaysia, Scotland, and Uruguay. HIV-1 subtype A is predominant in regions of Eastern Africa, Asia, and other former Soviet Union countries (Aibekova et al. 2018). As describe earlier, HIV-1 subtype B predominates in regions of North America, South America, Australia, Europe, Northern Asia, Middle East, and Northern Africa (Sallam et al. 2017). Subtype F is mainly in regions of Eastern Europe and southern America, while subtypes E, G, H, J, and K are mainly in countries of Central and West Africa (Bbosa, Kaleebu and Ssemwanga 2019) (Fig. 2). In cases of dual or super infection by two different HIV-1 strains within an individual person, two HIV-1 strains can simultaneously infect cells leading to recombinant HIV-1 with chimeric genomes. Co-circulation of different HIV-1 subtypes in the same geographical region result in unique recombinant forms, and if new transmission events occur, stable circulating recombinant forms (CRFs) can emerge, resulting in 'jumps' of HIV-1 genetic diversity (Sallam et al. 2017). Approximately 20% of group M viruses found in the HIV-1 epidemic are not 'pure' subtypes, rather they are

recombinant forms with 101 CRFs identified as of 2020 (<https://www.HIV-1.lanl.gov/content/sequence/HIV-1/CRFs/CRFs.html>).

HIV-1 subtypes and CRFs differ by 18% amino acid diversity (CI:15.6%–19.6%) on average and 12% (CI:8.6%–14.4%) between HIV-1 isolates within a subtype cluster, and 1% (CI:0.3%–2.2%) amino acid sequence diversity within the swarm of HIV-1 strains within an infected individual not receiving treatment (Li et al. 2015). The HIV-1 *env* gene shows the greatest genetic distance (35% median amino acid difference) between subtypes and up to 20% difference within a subtype (Korber et al. 2001, Gaschen et al. 2002). On the other hand, the *gag* and *pol* genes are more conserved between subtypes, demonstrating 17% and 6% amino acid differences on average, respectively (Korber et al. 2001, Li et al. 2015). Within the *pol* region target by nearly all antiretroviral drugs, HIV-1 integrase (IN) is more conserved compared to reverse transcriptase (RT) and protease (PR), with amino acid diversity of ~7% between subtypes and ~5% mean intra-subtype diversity (Rhee et al. 2008). These genetic differences in the PR, RT, and IN can lead to reduced or hypersusceptibility to antiretroviral inhibitors targeting these enzymes/enzymatic activities.

The majority of approved cART regimens have been developed based on the virology of subtype B HIV-1, which circulates in developed nations/regions and accounts for only 10% of global HIV-1 infections (Geretti 2006, Hemelaar et al. 2006a; Bbosa, Kaleebu and Ssemwanga 2019) (Fig. 2). Fortunately, these cART regimens are also effective at inhibiting HIV-1 non-B isolates in tissue culture and in PLWH (Darbyshire and Aboulker 1996, Hammer et al. 1996, Katlama 1996, Schooley et al. 1996). In a study of 2116 PLWH in the UK Collaborative Group on HIV Drug Resistance and the UK Collaborative HIV Cohort Study databases, PLWH infected with HIV-1 subtypes A, D, C, and CRF02_AG receiving mainly

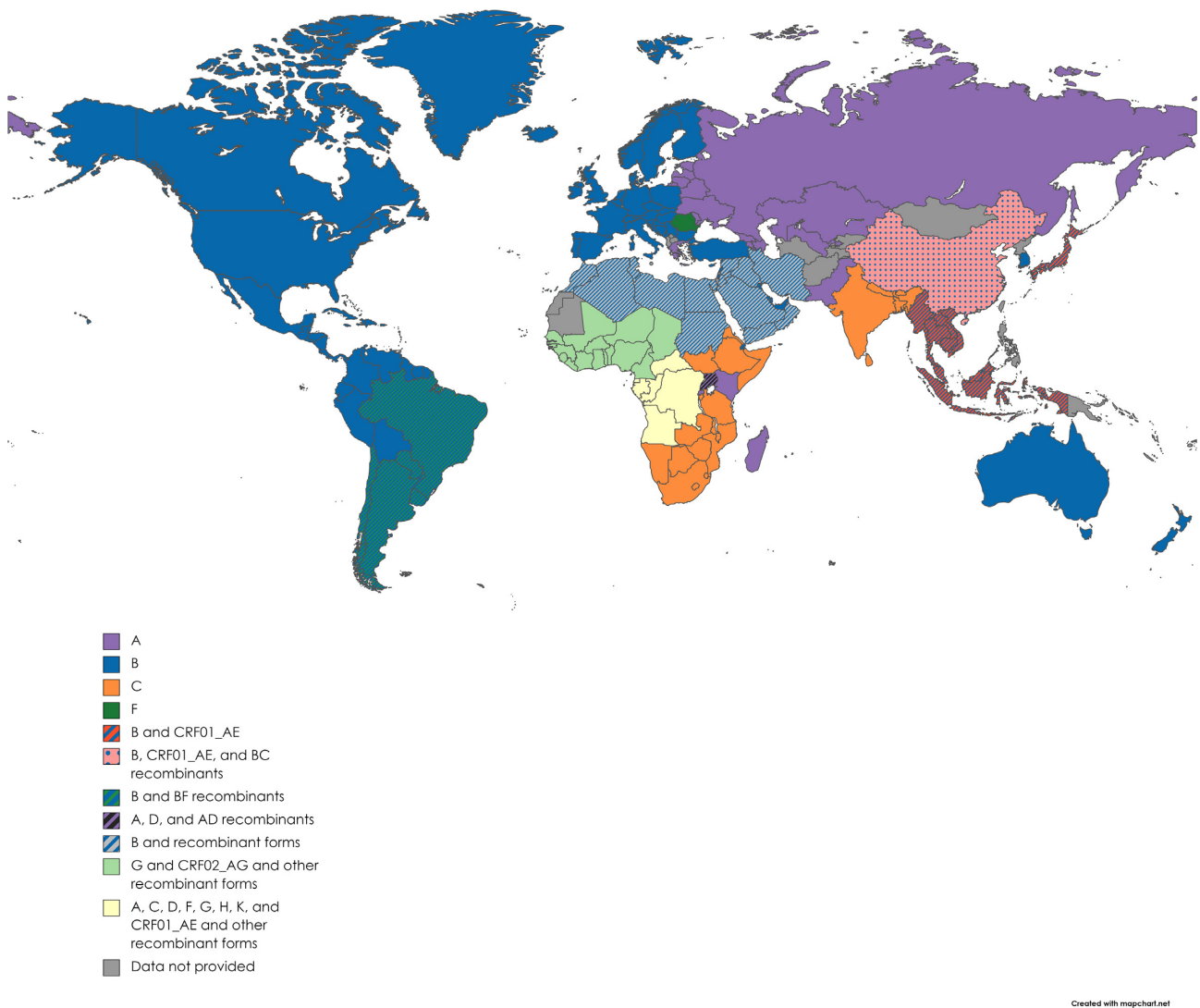


Figure 2. Distribution of dominant HIV-1 subtypes by region. The color presentation of different HIV-1 subtypes is embedded in the figure.

AZT + 3TC + efavirenz (EFV) had viral load suppression and CD4 cell count recovery comparable to individuals infected with HIV-1 subtype B virus (Geretti et al. 2009). However, a study of 861 HIV-1 subtypes B, C, D, A, and CRF02_AG infected individuals showed a faster CD4⁺ T cell decline and a higher frequency of virological failure among HIV-1 subtype D infected individuals (Easterbrook et al. 2010). There are a variety of socioeconomic factors leading to reduced cART adherence that relate to this difference in cART suppression whereas virus subtype has been considered a minor factor. However, when human conditions and socioeconomic factors are removed from analyses (which is nearly impossible), there remains the confounding factor of possible differential response to cART related to infection by highly divergent HIV-1 isolate/subtypes distributed in different regions around the world.

Rollout of cART in LMICs

Globally, access to cART has increased tremendously since early 2000s, such that by the end of 2019, over 26 million people had access to treatments, of which 17.9 million (69%) were PLWH in sub-Saharan Africa (UNAIDS 2020). In low- to middle-income countries (LMICs), treatment guidelines, including those provided by the World Health Organization (WHO), aim to max-

imize societal benefit within a resource-limited context (WHO 2016, NIH 2019). Nonetheless, the newest and most effective therapies remain protected by patents and are not affordable to these countries. Treatment options in LMICs are usually limited by the nation's ability to acquire generic versions of antiretroviral drugs. Historically, the first-line regimens in sub-Saharan Africa have included a triple regimen therapy of two NRTIs and one non-nucleoside reverse transcriptase inhibitor (NNRTI), commonly EFV or NVP (HIV/AIDS Programme 2006). The NRTIs didanosine, abacavir (ABC), lamivudine (3TC), and thymidine analogues AZT and d4T were commonly used in three-regimen cART for PLWH initiating treatment from the late 1990s to at least 2010 in these low resource settings but didanosine, d4T, AZT, and NVP were seldomly employed for treatment in high income countries during this time period. The latter regimens led to long-term toxicity, particularly lipodystrophy or fat-wasting as well as an accumulation of virological failure observed in 15%-35% of people living with HIV-1 in sub-Saharan Africa. Thus, WHO recommended the substitution of thymidine analogues for the preferred NRTI backbone of tenofovir (TDF) + 3TC (or emtricitabine; FTC) in adults or adolescents, and ABC + 3TC in children aged 3–10 (Gupta et al. 2009, Hoffmann et al. 2009, WHO 2015).

Following the 2016 approval of generic DTG for use in LMICs (CLINTON Health Access Initiative 2016), WHO revised their cART guidelines in 2019 to strongly recommend DTG + 2NRTIs as the preferred initial treatment in all populations with EFV 400 mg + 2 NRTIs only as the alternative regimen (aidsmap 2019, WHO 2019a, 2020). This replaced previous WHO guidelines which recommended EFV-based regimens as preferred first line cART (WHO 2016, CLINTON Health Access Initiative 2017). Nonetheless, generic DTG was slow to be implemented, with only slightly over one third of PLWH in LMICs having access to generic DTG by 2019 (Lancet HIV 2020). During this time, only a few LMICs such as Brazil and Botswana had begun providing DTG at the national scale. As of 2020, a more rapid rollout had begun in many, but not in all LMICs (Fig. 3A). In contrast, the penetration of DTG into cART regimens in high income countries (HICs) was rapid and starting soon after its FDA approval in 2013. Use of DTG in LMICs has continued to increase in recent years while first-generation NNRTIs, EFV, and NVP are phased out (Fig. 3B).

First-line TDF-based backbones have now replaced the use of AZT, which is now employed in second-line therapies. Countries transitioned away from d4T following its demotion in the 2010 WHO guidelines (CLINTON Health Access Initiative 2013). However, AZT and/or D4T were the most prescribed ARVs in the world until ~2012 despite the near absence of their use in HICs due to significant adverse/side effects such as anemia, lactic acidosis, peripheral neuropathies, and lipodystrophy. TDF became more widely used circa 2012 in part from improvements in process chemistry and lower-cost sources of raw material and because of voluntary licenses for generic versions of these drugs. TDF plus 3TC or FTC backbones are currently recommended by WHO in first-line treatments (WHO 2013) and TAF (a prodrug of TDF) use is expected to increase if ongoing research continues to support its use (Fig. 3C). While most high-income countries have access to a larger arsenal of second-line therapy options, LMICs are usually restricted to regimens based on the protease inhibitors (PIs) atazanavir and lopinavir (Fig. 4D), with DTG now being recommended for new second-line treatments in many countries when first line was NNRTI + 2 NRTIs (CLINTON Health Access Initiative 2020).

HIV-1 drug resistance presents a huge threat to viral suppression in PLWH on current antiretrovirals, and HIV-1 subtype may influence cART outcomes and the emergence of HIV-1 drug resistance (Ndashimye 2017, Sallam et al. 2017, Ndashimye and Arts 2019). The following sections described the differential response of HIV-1 subtypes to inhibition by various antiretrovirals as well as different evolutionary paths to drug resistance. We provide the literature-based evidence for the critical need to assess novel HIV-1 inhibitors within the context of diverse non-subtype B HIV-1 prior to approval and widespread use.

HIV-1 subtype specific resistance to the current cART

The currently available cART regimens were mainly developed by pharmaceutical companies based in high-income countries where HIV-1 subtype B predominates among infected individuals (Geretti 2006, Hemelaar et al. 2006, Bbosa, Kaleebu and Ssemwanga 2019). However, HIV-1 non-B subtype accounts for 90% of global infections (Snoeck et al. 2006) (Hemelaar et al. 2006). Individuals infected with non-subtype B HIV-1 could have sub-optimal responses to certain ARVs. The extreme scenario is intrinsic NNRTI resistant observed with HIV-2 and HIV-1 group O strains (Descamps et al. 1997, Tuillon et al. 2004). Ultimately, even

sub-optimal inhibition of non-subtype B HIV-1 coupled with poor adherence may result in disproportional prevalence of HIV-1 drug resistance in some HIV-1 subtypes, and thus, in some regions of the world. The following section describes variable sensitivity, intrinsic resistance, and differential rates of acquired resistance to antiretrovirals related to specific HIV-1 subtypes.

Subtype-specific resistance mutations to NRTIs and NNRTIs

Generally, amino acid substitutions conferring resistance to NRTIs in subtype B match those of non-subtype B (Bellocchi et al. 2005). Mutation of lysine 65 of HIV-1 RT confers resistance to TDF, ABC, didanosine, and d4T in most HIV-1 subtypes. However, there is more rapid emergence of K65R mutation in the RT coding region in PLWH infected with subtype C compared to individuals infected with subtypes B, G, and CRF02_AG (Brenner et al. 2006). Lysine 65 is encoded by AAG in subtype C and AAA in subtype B such that a single versus double A to G transition mutation (respectively) is necessary for the arginine mutation (encoded by AGG) at this position. On the contrary, K65R is less frequent in subtype A compared to subtypes B and C HIV-1 in PLWH on cART which is not fully understood (Gupta et al. 2005). The frequency of thymidine analog (d4T or AZT) drug resistance mutations (D67N, K70R, L210W, and T215F/Y) is similar in PLWH infected with HIV-1 subtypes A, B, and C, although the thymidine analog resistance-conferring mutation M41 L is more frequent in subtype B than C infected individuals when receiving cART with a thymidine analog (Soares et al. 2003). M41 L is less likely to be selected by subtype CRF02_AG (adjusted odds ratio (AOR) = 0.35; $P = 0.022$) than subtypes CRF06_cpx, G, or A (Chaplin et al. 2011). Also, the thymidine analog resistance mutation L210W may have increased prevalence in subtype A (AOR = 42.5, $P = 0.001$) infected individuals compared to subtype CRF02_AG, CRF06_cpx, and G viruses (Chaplin et al. 2011).

In addition to these approved RT inhibitors, there are several NRTI and NNRTIs currently in clinical trials which have strong cross-subtype inhibition but poorly characterized resistance profiles in non-subtype B HIV-1. For example, GS-9131 (a prodrug of GS-9148) is a deoxyadenosine analog that potently inhibits HIV-1 subtypes A, B, C, D, E, F, and group O and N viruses ($IC_{50} = 0.29\text{--}48.1\text{ nM}$). GS-9131 can also inhibit HIV-1 carrying NRTI resistance-associated mutations K65R, M184V, and L74V (White et al. 2017). Currently in development by Merck and Co, islatravir (MK-8591) is a potent, long-acting NRTI against HIV-1 and HIV-2, including those isolates carrying the M184V mutation ($IC_{50} \leq 11\text{ nM}$) (Wu et al. 2017). Furthermore, adding K65R, Q151M, or both mutations to viruses carrying M184V does not seem to increase resistance to MK-8591 compared to the M184V mutation alone (Wu et al. 2017). The activity of islatravir against viruses carrying M184V may be of particular importance to individuals infected with resistant virus to NNRTIs and the T-20 inhibitor.

In relation to NNRTI resistance, mutations in HIV-1 RT associated with NVP and EFV resistance, namely K103N, V106M, and A98S, occur more frequently in those infected with HIV-1 subtype C compared to subtype B. This was observed in both cART naïve and experienced PLWH receiving an NNRTI-based treatment regimen (Grossman et al. 2004). K103N is also less frequent in HIV-1 subtypes A and D compared to subtype C (Eshleman et al. 2005). Y181C, which reduces susceptibility to NVP, EFV, etravirine, and rilpivirine, shows greater prevalence in subtypes B and C compared to subtype A HIV-1 (Gupta et al. 2005). Minority species of K103N or Y181C mutant viruses detected by ultrasensitive PCR were found in 42% of subtypes A and CRF01_AE compared to 70–

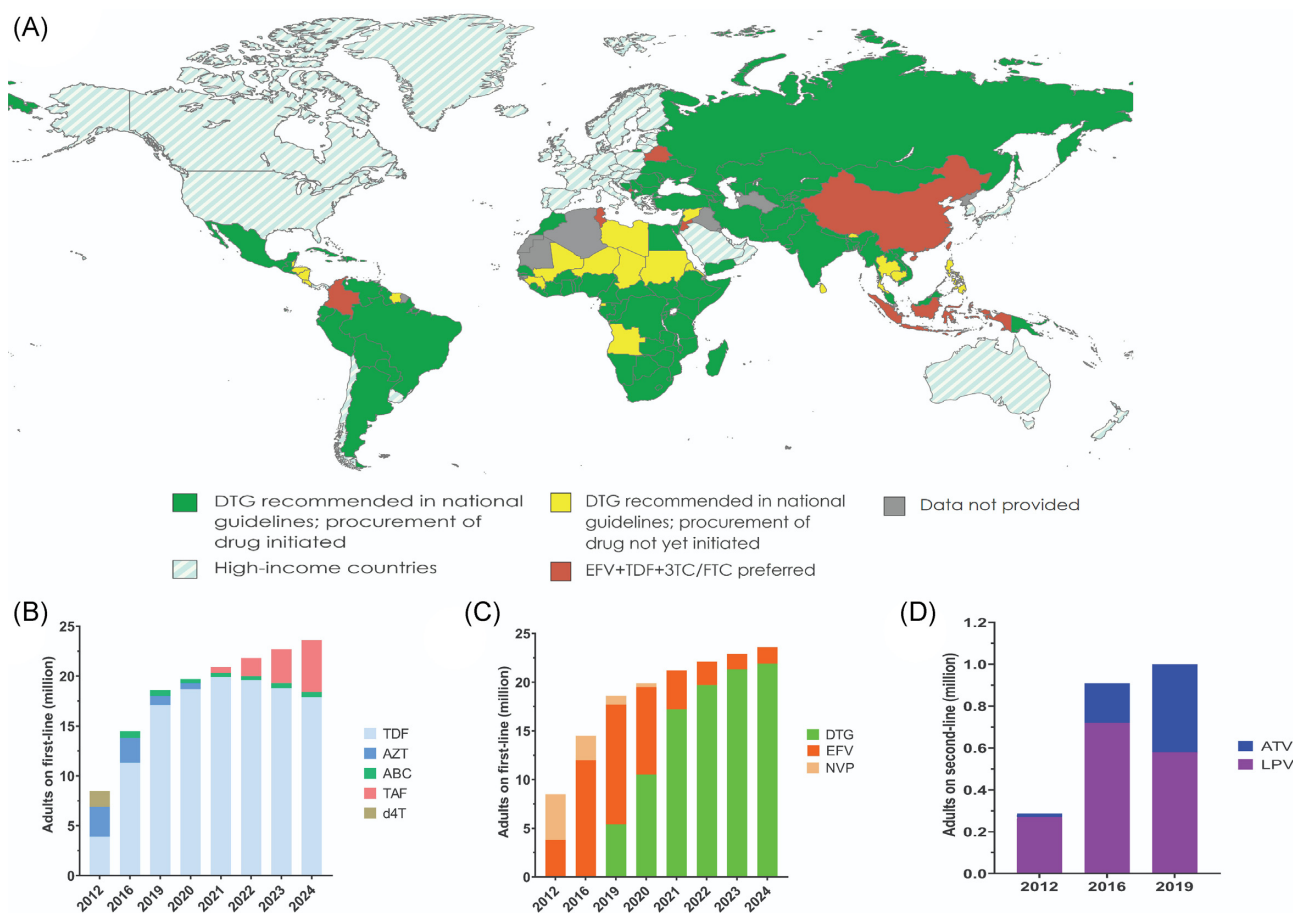


Figure 3. Adult antiretroviral use in first- and second-line therapies in LMICs. **(A)** Status of dolutegravir uptake in LMICs as of July 2020. **(B)** Estimated use of NNRTI/INSTIs and **(C)** NRTIs as components of adult first-line therapies in generic accessible-LMICs. **(D)** Use of PIs as components of adult second-line therapies in LMICs.

87% of subtype C HIV-1 infections (Brenner 2007). While V106M is preferentially selected in subtype C under NVP/EFV therapy, V106A also confers resistance to NVP and EFV and is commonly seen in subtype B viruses (Loomba et al. 2002, Brenner et al. 2003). N348I reduces susceptibility to NVP, EFV, AZT, and didanosine and is commonly associated with M184V and thymidine analog mutations in PLWH (Nikolenko et al. 2007, Yap et al. 2007, Hachiya et al. 2008, von Wyl et al. 2010). There seems to be less susceptibility for EFV and NVP in subtype A and D viruses than for subtype B and C when the N348I mutation is present, and this mutation has frequently been observed in HIV-1 subtype C viruses (Brehm et al. 2012).

Rilpivirine is a second generation NNRTI with a high genetic barrier to resistance *in vitro*, but with reduced susceptibility in the presence of E138A/K/Q mutations (Anta et al. 2013). In cART naïve and experienced PLWH, E138A/K/Q mutations are found at a higher frequency in HIV-1 subtype C than subtype B infected individuals (Sluis-Cremer et al. 2014). NVP-, EFV-, and doravirine-associated mutations A98G (AOR = 0.48, $P = 0.019$) and V106I (AOR = 3.16, $P = 0.003$) have been associated with increased risk in subtype G infected individuals relative to CRF02_AG, CRF06_cpx, and subtype A (Chaplin et al. 2011). V90I, a weakly selected mutation by NVP and EFV, has shown increased emergence in subtype CRF02_AG (AOR = 3.16, $P = 0.003$) infected individuals failing first-line therapy in Nigeria (Chaplin et al. 2011). Finally, a highly resistant subtype D HIV-1 strain was isolated from a treatment naïve Ugandan patient with a combination of I135L, T139V, and

V245T; these mutations had limited impact on replicative fitness when present in this subtype and may represent a pathway for NNRTI resistance unique to clade D (Gao et al. 2004). Interestingly nearly all NNRTI and NRTI mutations are conferred by mutations impacting the function of the catalytically active subunit of RT, p66 in the p66/p51 heterodimer (Arts et al. 1996, Huang et al. 1998). The p51 shares the same sequence as p66 except for the 15 kDa C-term deletion but is folded in alternative conformation for structure support of p66. Mutations at amino acid positions at 135, 138, 139 and 245 are found in the p51 subunit of the NNRTI binding pocket in RT and confer resistance to NNRTI. Resistance conferring mutations at these sites are also more common in non-subtype B and as such are often under-reported.

Subtype-specific resistance mutations to PIs

In most sub-Saharan African countries, due to breaks in drug supply/access and the low genetic barrier to HIV-1 drug resistance mutations associated with first line NRTI + NNRTI regimens (WHO 2019b), 10%–15% of PLWH who start cART fail within one year, and 70%–80% of people with virological failure develop acquired drug resistance (Clutter et al. 2016). These regimens frequently prescribed to PLWH in LMICs also have high pill burden which results into poor adherence accelerating the rates of treatment failure and development of drug resistance. Previously, a second line cART in LMICs consisted of 2 NRTIs + protease inhibitor (commonly lopinavir/ritonavir or atazanavir/ritonavir)

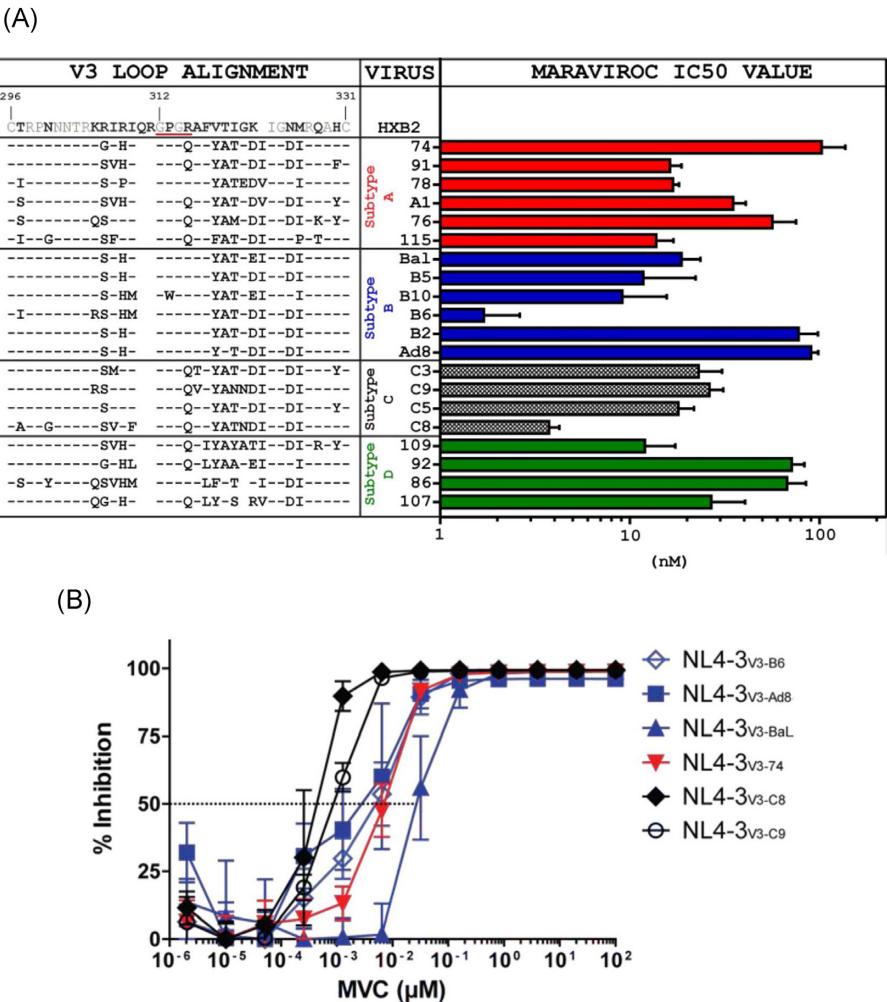


Figure 4. Sensitivity of HIV-1 Primary Isolates to Maraviroc. **(A)** IC50 values of a panel of HIV-1 group M viruses from primary isolates including subtype A (red), subtype B (blue), subtype C (grey), and subtype D (green) to maraviroc were determined in U87.CD4.CCR5 cells at a multiplicity of infection of 0.01 IU/cell. Alignments of the V3 loop region of gp120 with reference virus HXB2 are also shown. **(B)** Drug sensitivity curves of V3 loop chimeric NL4-3 viruses generated from primary isolates in Fig. 4A were performed in U87.CD4.CCR5 cells. Data represents triplicates with standard deviations shown.

(Fig. 3D). Currently, DTG plus optimized 2 NRTIs is the preferred second-line treatment in adults and children failing on non-DTG regimens, while lopinavir/ritonavir or atazanavir/ritonavir are used in PLWH failing on DTG or EFV-based first-line cART (WHO 2018). The choice of NRTI backbone to use in second-line therapy depends on which NRTI was used in the first line. If ABC + 3TC or TDF + 3TC (or FTC) were used, AZT + 3TC is commonly used in second-line therapy. Likewise, 3TC with either ABC or TDF is preferred for PLWH failing on AZT-containing first-line regimens (WHO 2018).

Natural polymorphisms L10I, L10V, K20R, M36I, L63P, V77I, and V82I found in the protease gene of non-subtype B HIV-1 are known to reduce susceptibility to PIs in HIV-1 subtype B viruses (Grossman et al. 2001, Holguín et al. 2004). Substitutions at position 36 of HIV-1 protease seem to have a differential impact on susceptibility to treatment in different subtypes. M36I appears as an accessory PI resistance mutation (Lisovsky et al. 2010) and enhances resistance to ritonavir, atazanavir, and nelfinavir as a natural polymorphism in non-B HIV-1 subtypes, specifically subtype C and CRF02_AG (83%), but appears at only a 6% occurrence in subtype B cART naïve individuals (Pieniazek et al. 2000).

Primary drug resistance mutation V82A/F/T/S selected by indinavir and lopinavir appears more frequently in subtype F1 than subtype B HIV-1 infected individuals (Soares et al. 2003). In contrast, D30N results in highly impaired viral replication in HIV-1 subtype C, whereas the same mutation is a primary PI resistance mutation in HIV-1 subtype B infected individuals receiving a PI-based treatment (Gonzalez et al. 2004). A lack of D30N in some subtypes may relate to a lack of the natural polymorphisms L63P and V77I in subtype B compared to non-B subtypes that are necessary to compensate for fitness loss incurred by D30N (Pieniazek et al. 2000). L63P has also been shown to compensate for the loss in viral fitness caused by the primary PI resistance mutation L90M found at a lower frequency among subtypes C, F1, and intersubtype BF compared to subtype B (Carobene et al. 2004, Soares et al. 2007, Dumans et al. 2009). Finally, emergence of M89T, which causes phenotypic resistance to atazanavir, nelfinavir, and lopinavir, is influenced by the presence of the polymorphism M89 in subtypes A, C, and CRF01_AE, and L89 in subtype B viruses (Martinez-Cajas et al. 2012). This leads to preferential selection of the M89T mutation in subtype C viruses (Martinez-Cajas et al. 2012). The

mutation L89M increases the genetic barrier for emergence of L90M and reduces susceptibility to indinavir, ritonavir and nelfinavir in subtype F, though susceptibility to these antiretrovirals is maintained in subtype B viruses, with the exception of indinavir (Calazans et al. 2005).

In-vitro assays also reveal possible polymorphisms associated with hypersusceptibility to PIs. For example, CRF02_AG viruses carrying protease polymorphisms G17E and I64M confer hypersusceptibility to nelfinavir, atazanavir, and indinavir inhibition (Santos et al. 2012). The polymorphism I93 L signature to HIV-1 subtype C is also associated with hypersusceptibility to lopinavir in subtype C viruses compared to HIV-1 subtype B viruses (Gonzalez et al. 2003). A significant increase in protease inhibitor (PI) resistance in subtype B vs. C infected individuals has also been observed (54% vs. 23%) (Grossman et al. 2001).

Subtype-specific resistance mutations to coreceptor and fusion inhibitors

In the late 1990s, screening compounds that block HIV-1 host cell entry was a promising strategy for the development of a new drug class and for therapeutic combination with nucleoside analogs. However, the use of these inhibitors was plagued by variable susceptibility among HIV-1 isolates of both B and non-B subtypes. The gp120-gp41 envelope glycoprotein spike (Env) binds to the cellular receptor (CD4) and then to one of two different coreceptor (CCR5 or CXCR4) facilitate HIV-1 entry and are thus important therapeutic targets. Fusion inhibitors block HIV-1 fusion with the host target cell, while coreceptor antagonists prevent the interaction between gp120 and the CCR5 or CXCR4 coreceptor. The first FDA-approved entry inhibitor, and only approved fusion inhibitor to date, is enfuvirtide (T20). This inhibitor is a synthetic peptide of 36-amino acids, corresponding to C-terminal residues 127–162 of subtype B gp41 heptad repeat (HR) 2, that prevents the formation of the 6-helix conformation required for membrane fusion and entry (Derdeyn et al. 2001). However, T20 has a short half-life *in vivo*, is poorly tolerated, and has a low barrier to resistance; consequently, it is used infrequently in cART regimens. Resistance to T20 occurs commonly in amino acid regions 36–45 of gp41 heptad repeat domain 1 and commonly includes the gp41 mutations G36D/S, V38A/M, Q40M, N42S/T/D/E, N43D/K/S, L44M, and L45M (HXB2 numbering). These mutations cause reduced susceptibility by 2–21-fold (Greenberg and Cammack 2004). Despite the high conservation of region 36–45 of gp41 in T20 naïve individuals, natural resistance mutations have been shown to occur more frequently in non-B HIV-1 subtypes (16.7%) compared to subtype B viruses (9.1%) (Carmona et al. 2005). HIV-1 subtypes A, B, G, and C may favour emergence of the N42S mutation associated with increased susceptibility to T20 compared to subtype F viruses (Carmona et al. 2005, Melby et al. 2006). In T20 naïve individuals, mutations L44M and G36S, are common natural resistance mutations in subtype B viruses compared to subtypes A, G, C or CRF02_AG and these mutations reduce T20 susceptibility by 2-fold and 7-fold, respectively (Carmona et al. 2005).

CCR5 antagonists bind to the hydrophobic pocket formed by transmembrane helices of CCR5 (Murga et al. 2006). This allosteric interaction results in locking of the receptor in a conformation unrecognisable by HIV-1 Env (Watson et al. 2005). Maraviroc (MVC), the only FDA approved CCR5 inhibitor, is recommended in treatment experienced PLWH infected with CCR5 using HIV-1 (Flexner 2007, Lalezari et al. 2007). Resistance to CCR5 antagonists mainly occurs through 3 mechanisms: (i) ability of HIV-1 Env to bind CCR5 bound to MVC (Kitrinos et al. 2009), (ii) selection of CXCR4 by orig-

inally CCR5 viruses (Connor et al. 1997, Mosier et al. 1999, Regoes and Bonhoeffer 2005), and (iii) increased affinity of CD4, allowing HIV-1 to remain bound to the cell surface in a conformation that engages CCR5 when MVC dissociates (Ratcliff, Shi and Arts 2013). Most mutations that confer resistance against MVC occur in the V3 loop of gp120, which was demonstrated by serial passage assays with escalating drug pressure resulting in selection of e.g. resistance mutations T316A and V323I in this V3 loop. Only one study has selected for MVC resistance from *de novo* mutations conferring a switch from a CCR5 to a CXCR4 using HIV-1 (Mosier et al. 1999). The percentage of people with CXCR4 using virus varies at different stages of HIV-1 infection. Typically, the CXCR4 using HIV-1 can reach up to 12–19% in recently diagnosed HIV-1 patients (Chalmet et al. 2012) but is typically found at low frequencies, often less than 1% of the CCR5-using HIV-1 in any given PLWH during asymptomatic disease (Frange et al. 2010). Upon treatment failure with MVC-containing cART, patients may carry CXCR4, CCR5 or mixed CXCR4/CCR5 virus population (Jubb et al. 2019), which is often associated with the pre-treatment detection of a very low frequency CXCR4 variant in the inpatient quasiespecies, often only detectable by next generation sequencing (NGS) and subsequent predictive algorithms for CXCR4 identification (Archer et al. 2009). Unfortunately, pre-screening with phenotyping or the aforementioned genotyping assays are not available or rarely used prior to treatment with MVC in low resource settings. Interestingly the removal of MVC from the treatment may result in a switch back to a dominant CCR5-using HIV-1 population in patients with a pre-existing mixed population of CXCR4 and CCR5 viruses at the initiation of therapy (Westby et al. 2006). Although MVC is rarely used for treatment in high income countries, this drug has been widely marketed in LMICs for inclusion in last cART alternatives. However, CXCR4-usage is more common in subtype D (Tscherning et al. 1998, Huang et al. 2007, Church et al. 2008) than any other HIV-1 subtype but is quite rare in subtype C HIV-1 (Tscherning et al. 1998, Ndung'u et al. 2006). Genotypic or phenotypic assays to screen for co-receptor usage are rarely used in LMICs for MVC treatment, despite this screen being an FDA requirement for including MVC in cART (Clinical INFO HIV gov 2020).

MVC resistance emerging through enhanced CD4 interactions is conferred by the N425K mutation in the CD4 binding pocket in gp120. This mutation provides a gain in replicative fitness, but never emerges in subtype B infected individuals with or without MVC treatment possibly due to enhanced inhibition by neutralizing antibodies (Ratcliff, Shi and Arts 2013). However, the N425K mutation is more frequent in the subtype A population than in all other subtypes. This suggests that if MVC was more widely employed for treatment, emergence of resistance may be more rapid in subtype A infected individuals.

Finally, IC₅₀ values may range $\pm$ 10% with most classes of drugs when comparing susceptibility of wild type HIV-1 of any subtype derived treatment naïve individuals. In contrast the IC₅₀ values to MVC and T20 (as attachment inhibitors, see above) can vary 100-fold when testing divergent HIV-1 isolates, typically related to envelope sequence difference between HIV-1 subtypes and typically this variable susceptibility is not mapped to specific drug resistance mutations/polymorphisms. This observation has been somewhat debated with (Dorr et al. 2005) who has shown similar MVC inhibition (geometric mean 90% IC of 2.0 nM, (95 CI, 1.8–2.4 nM) with diverse subtype B and non-B subtypes primary isolates (subtype A, C, D, E, G, J, and group O) from cART naïve and experienced PLWH (Dorr et al. 2005). In contrast, a series of studies (Torre et al. 2000, Lobritz, Ratcliff and Arts 2010, Ratcliff,

Shi and Arts (2013) showed differential sensitivity to MVC, PSC-RANTES, T20, and other entry inhibitors among HIV-1 of different subtypes. An example of this is with MVC and four HIV-1 viruses of four different subtypes (Fig. 4), which was published as background information in the Annette Ratcliff PhD Thesis (Ratcliff and Arts 2013). Finally, studies with recombinant viruses with patient derived HIV-1 envelopes showed that subtype C viruses were more sensitive to MVC compared to the Env of subtype B and recombinant CRF01_AE viruses ($P = 0.002$) (Siddik et al. 2018).

Subtype-specific resistance to INSTIs

HIV-1 integrase strand transfer inhibitors (INSTIs) target HIV-1 integrase to stop viral replication. INSTIs are the most recent, potent class of antiretroviral drugs but resistance to these inhibitors in non-subtype B HIV-1 has been poorly studied due to their slow rollout in LMICs. As such, this review is more focused on INSTIs and emerging data on INSTI resistance in LMICs.

Integrase catalyses HIV-1 integration through 3' endonucleolytic processing of viral DNA ends in the cytoplasm and strand transfer in the cell nucleus where viral DNA is covalently linked to cellular DNA. For the INSTIs to be active pharmacophores, they must have a chemical group of three oxygen atoms in the same geometric plane and a halogenated phenyl group in their chemical structure. The halobenzyl moiety fits within a tight catalytic pocket of HIV-1 integrase that is made available after 3' processing. Here, it interacts with the G:C base pair preceding the terminal adenine of the 3' HIV-1 LTR, as well as with residues 145 and 146 of HIV-1 integrase. These interactions result in the displacement of the reactive adenine and the 3' hydroxy group from the active site, inactivating the intasome and stabilising the HIV-1 integrase-viral DNA complex (Hare et al. 2010a,b). The three oxygen atoms of the β -hydroxy ketone moiety chelates divalent cations within the catalytic pocket, impeding the catalytic triad D-D-E from participating in the strand-transfer process (Grobler et al. 2002, Bacchi et al. 2011a,b, Engelman and Cherepanov 2012). The length of the linker between the halobenzyl group and coplanar oxygen atoms distinguishes the types of INSTIs currently in use. Whereas the first generation INSTIs like raltegravir (RAL) and elvitegravir (EVG) have a short linker, the linker in the second generation INSTIs, such as DTG, bictegravir (BIC), and cabotegravir (CAB), is further extended and enables deeper access into the pocket vacated by displaced terminal adenine to allow a closer interaction with viral DNA. Despite BIC and CAB being structurally and chemically similar to DTG due to the presence of a tri-cyclic central pharmacophore, the metal-chelating scaffold for CAB is 5-membered (more rigid) compared to the more flexible 6-membered ring in DTG (Fig. 5). Overall, BIC seems to be more effective than either CAB or DTG, which may be explained by further flexibility of the oxazepine ring that allows accommodation to structural changes of the integrase active site in the presence of resistance mutations (Smith et al. 2018). This structural difference partly explains the low genetic barrier to drug resistance associated with RAL and EVG compared to the second generation INSTIs.

RAL is only sparingly used in LMICs despite its approval by the FDA for use in both cART naïve and experienced PLWH in 2007. Rather, RAL is commonly used as a salvage regimen in combination with darunavir boosted with ritonavir and etravirine in heavily treated PLWH failing on NNRTI and PI regimens. DTG, which received FDA approval in 2013, has only recently become widely available in LMICs as of 2018 and is commonly used as first-line or second-line treatment in PLWH failing with no previous exposure to INSTIs (Fig. 3B). BIC was approved by the FDA in 2018 as a fixed-

dose combination of BIC/FTC/TAF in cART-naïve and -suppressed PLWH (<50 copies/mL) with no history of drug resistance. The FDA and the EU recently approved CAB as a long-acting injectable, cabotegravir/rilpivirine, for use in PLWH with undetectable viral loads who are currently stable on cART without drug resistance. However, these newly developed second-generation INSTI formulations are not widely available in LMICs (Gilead 2018; FDA 2021; GSK 2021; Simon Collins 2021).

Among HIV-1 subtypes, residues critical for integrase catalytic activity (D64, D116, and E152) and for zinc binding (H12, H16, C40, and C43) are fully conserved (variability $\leq 0.5\%$) (Ceccherini-Silberstein et al. 2009). The high degree of conservation (variability $\leq 0.25\%$) is also seen with HIV-1 integrase residues (H12, L102, A128, A129, C130, W131, W132, I161, R166, Q168, E170, T174, M178, and Q214) which interact with cellular cofactor human lens epithelium-derived growth factor (LEDGF/p75) responsible for tethering the HIV-1 preintegration complex to cellular chromatin (Hombrouck et al. 2007, Ceccherini-Silberstein et al. 2009). Upon INSTI treatment, resistance can emerge in primary mutation sites N155, Q148, and Y143 with all subtypes. However, even in untreated individuals, there is a degree of variability at secondary resistance mutations sites among different HIV-1 subtypes (Fig. 6) (Loizidou et al. 2009). HIV-1 subtype-specific amino acid differences have been observed near primary or at secondary resistance mutation sites (Myers and Pillay 2008).

Mutations at position G140 of HIV-1 integrase occur more frequently in subtype B than non-subtype B infected individuals (Doyle et al. 2015). G140A/S does not confer INSTI resistance on its own, but it increases the frequency of the primary Q148H/K/R resistance mutation in subtype B viruses (Table 1). The rare appearance of G140A/S upon INSTI treatment failure is attributed to a high genetic barrier with non-B subtypes compared to subtype B. In subtype B virus, glycine at position 140 is commonly encoded by the codons GGT or GGC. Therefore, a single G to A transition mutation at the first codon position replaces glycine with serine. In contrast, glycine is commonly encoded by either GGG or GGA in non-B subtypes and thus, two nucleotide substitutions (a transition and transversion) are required for a G140S mutation to arise. In cell culture assays, the R263K mutation is generally selected by DTG and EVG in HIV-1 subtype B, whereas R263K is rare and G118R is more common in CRF02_AG and C subtypes to confer resistance to DTG, RAL, and EVG (Quashie et al. 2012). Whereas G118R causes low-level resistance to RAL in subtypes B and C, there is increased RAL resistance by G118R in CRF02_AG in cell culture and patient samples (Malet et al. 2011, Quashie et al. 2014). In subtype C, the high replicative fitness cost of R263K likely offsets the level of resistance and thus, G118R is favored with a lower level of INSTI resistance.

The mutation E138K, which confers resistance to RAL, EVG, and DTG in the presence of Q148 mutations, shows increased resistance to RAL in CRF02_AG compared to subtype B viruses (Quashie et al. 2014) (Table 1). Double mutation E92Q/N155H in HIV-1 subtype C backbone shows 10-fold greater susceptibility to RAL and EVG compared to the same mutations in subtype B viruses (Bar-Magen et al. 2010). Subtype can also affect integrase enzyme activity: H51Y causes significant loss in integrase enzyme activity in CRF02_AG and B subtypes but not in subtype C. E138K leads to > 55% reduction in enzyme activity in CRF02_AG but not in subtypes B and C (Quashie et al. 2014). Strand transfer efficiency is restored by addition of E138K/H51Y to G118R in subtype B, although this occurs to a lesser extent in subtypes C and CRF02_AG (Quashie et al. 2014).

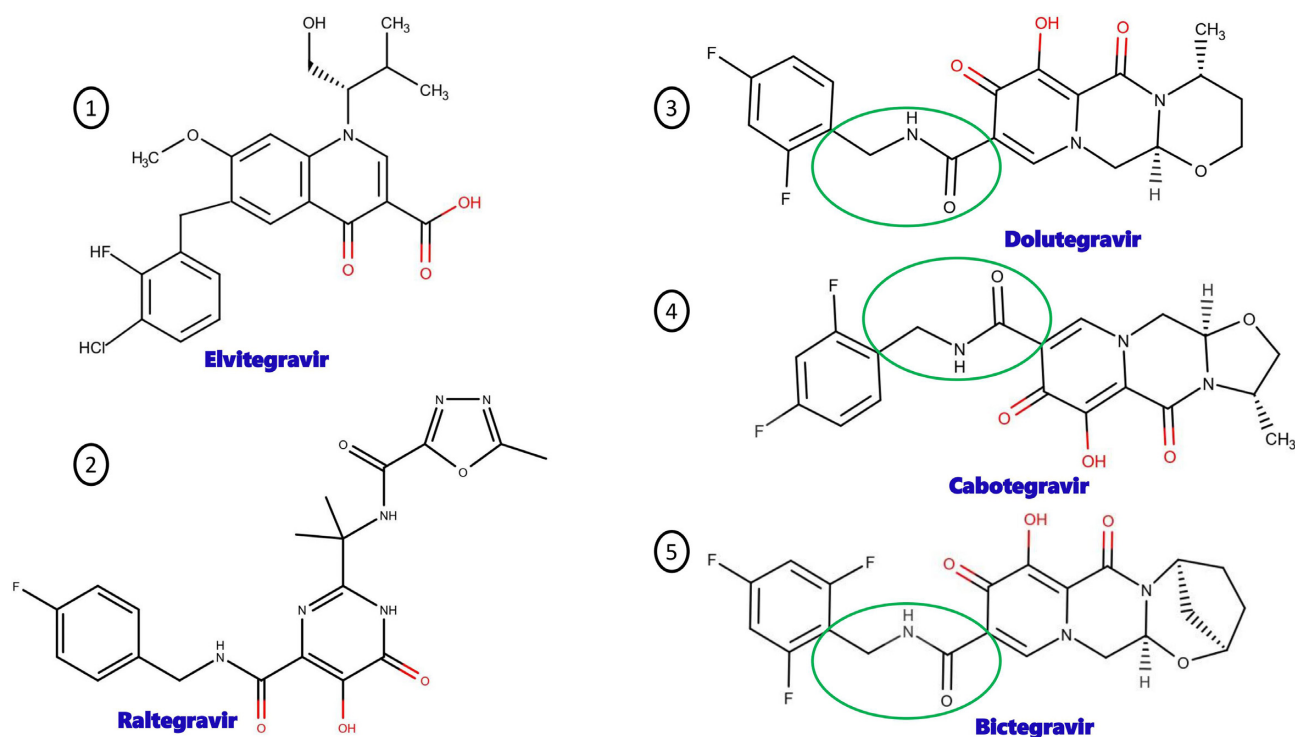


Figure 5. The chemical structures of approved INSTIs. The coplanar oxygen atoms are highlighted in red; green circles highlight extended linkers, which are a common feature of all second-generation INSTIs.

We have previously shown that Q148H/R/K mutations can occur singly and more frequently in subtype B viruses compared to non-B HIV-1 subtypes (Ndashimye et al. 2020, 2021). This may largely be due to the high frequency of G140S mutations in subtype B. On the contrary, in rare occurrences of Q148H/R/K mutations in non-B subtypes, they mainly occur in complex with other primary INSTI mutations, including E138K/A, G140A, S147G, and the accessory mutation T97A. The detection of Q148H/R/K mutations occurs predominantly in heavily treated PLWH who failed on NRTI-, NNRTI-, and PI-containing regimens (Ndashimye et al. 2018, 2020, 2021, Seatla et al. 2018, Achieng and Riedel 2019, Fokam et al. 2020). This may translate to better treatment outcomes with the use of second generation INSTIs in LMICs, especially if virological monitoring is widely accessible to detect early virological failure. Whereas G140S/Q148H variants are common in subtype B infected individuals failing on RAL, it is G140A/Q148K/R substitutions that predominate in subtype A, D, C, and CRF18_cpx viruses in PLWH failing on RAL-regimens (Ndashimye et al. 2018, 2021, Seatla et al. 2018, Achieng and Riedel 2019, Fokam et al. 2020).

HIV-1 subtype specific resistance to novel classes of HIV-1 inhibitors

HIV-1 attachment inhibitors

HIV-1 attachment inhibitors are a relatively new class of antiretrovirals which bind, modulate the conformation, and inhibit gp120 of the Env trimer, thereby preventing HIV-1 entry into host CD4⁺ T cells. Fostemsavir (BMS-663068), prodrug of temsavir (BMS-626529), was recently granted FDA approval and constitutes the first and only clinically approved HIV-1 attachment inhibitor with activity against CCR5 and CXCR4 usage viruses (Li et al. 2013b, FDA 2020, ViiV Healthcare 2020). The approval

was supported by phase III clinical trial BRIGHTER which tested for efficacy and safety of temsavir in individuals carrying multi-resistant HIV-1, with majority having advanced HIV-1 disease. In the study, 60% ($n = 163/272$) of PLWH on temsavir and optimised background regimens had < 40 HIV-1 RNA copies/ml through week 96 (Ackerman et al. 2019). Temsavir functions by binding a surface-accessible pocket in gp120 to stabilize the prefusion Env conformation and block CD4 receptor binding (Langley et al. 2015, Pancera et al. 2017). The IC_{50} of temsavir was determined against a large panel of subtype-diverse HIV-1 pseudoviruses, revealing values that ranged from < 1 to $> 20,000$ nM (Pancera et al. 2017). As such, temsavir is believed to have the greatest variability of inhibition of non-subtype B HIV-1 compared to all currently approved anti-HIV-1 drugs. Specifically, temsavir poorly suppressed all CRF01_AE viruses (23/23), 100-fold greater the average IC_{50} value compared to that with subtype B HIV-1 ($IC_{50} > 1000$ nM with CRF01, < 10 nM with subtype B). Fourteen of these 23 CRF01_AE isolates were completely resistant to temsavir inhibition ($IC_{50} > 20,000$ nM). Also, while temsavir potently inhibited most subtype B (29 of 41) and C (39 of 58) variants, this level of inhibition was achieved in only 7/27, 1/7, 1/5, 1/3, and 5/16 subtype A, G, AC recombinant, AD recombinant, and AG recombinant viruses, respectively. These results indicate that sensitivity to temsavir inhibition is highly variable across group M HIV-1, and significantly, this compound may not be suitable for individuals infected with non-subtype B isolates and many recombinant strains (in LMICs or HICs). While most residues within the temsavir binding pocket are highly conserved, gp120 positions M426, S375, M434, and M475 show some amino acid variability, with mutations at these positions contributing significantly to temsavir susceptibility (Zhou et al. 2014, Pancera et al. 2017). These variable residues lie within 5 Å distance from the temsavir binding pocket, with S375 lying even closer within 2.5 Å, which

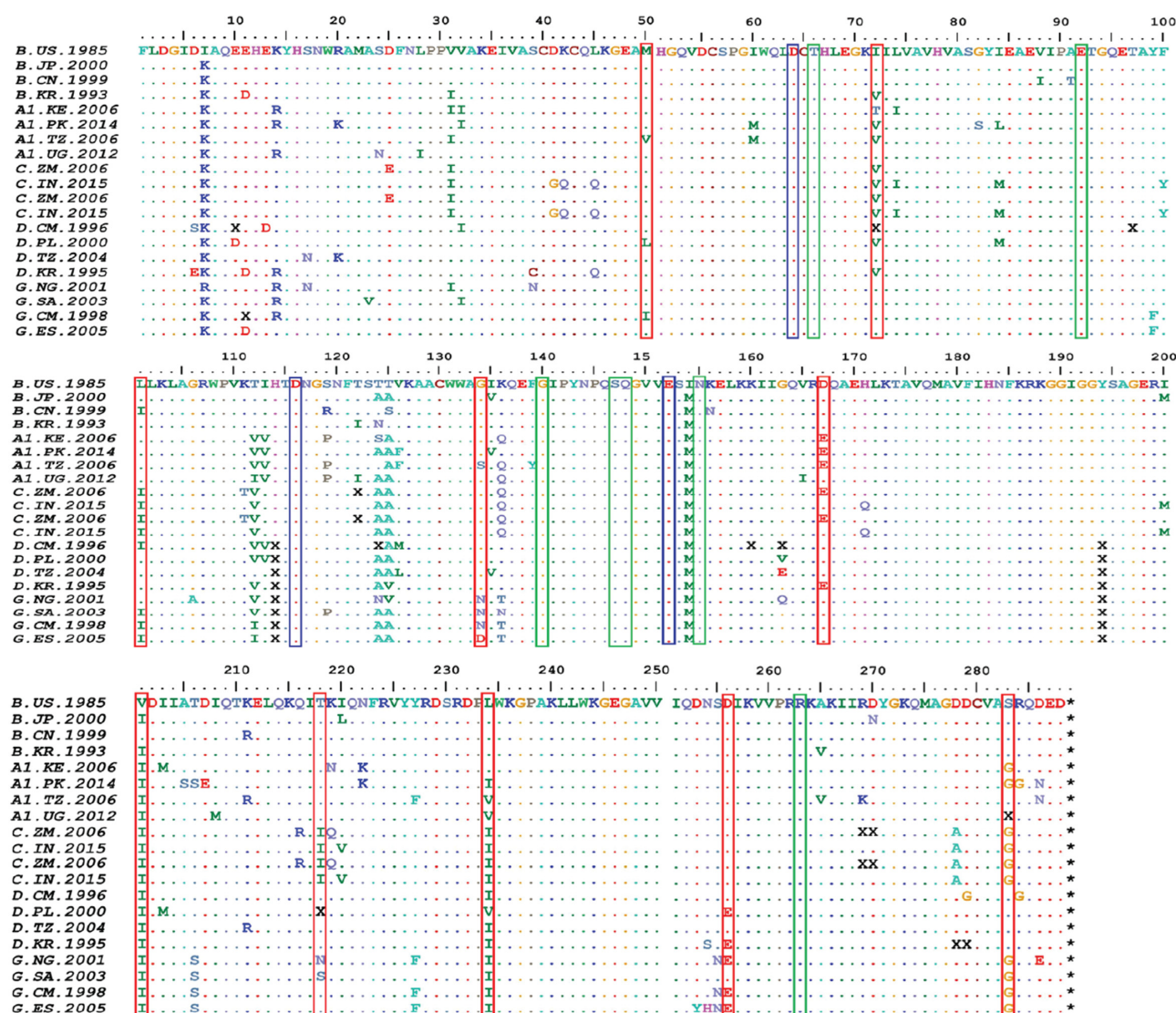


Figure 6. HIV-1 integrase variability among different subtypes. Across the HIV-1 integrase gene, there are several differences in naturally occurring polymorphisms in different HIV-1 subtypes. These polymorphisms may function as compensatory mutations in specific subtypes which may influence differential selection of major mutations. HIV-1 integrase sequences of cART naïve patients were downloaded from HIV-1 sequence database (<http://www.HIV-1.lanl.gov/>). Blue rectangular boxes show the three catalytic triad residues of CCD (D64, D116, and E152), green- major resistance mutation positions and red-polymorphisms naturally occurring between different HIV-1 subtypes. General consensus of amino acids at a particular position between HIV-1 subtypes is represented by (.) .

indicates potential strong covalent interactions with temsavir (Fig. 7).

Temsavir resistance mutations at positions, 375, 426, 434, and 475 of HIV-1 gp120 are natural polymorphisms in subtypes C and D, as well as in recombinants CRF02_AG and CRF01_AE (Fofana et al. 2015). M426 L has been found in subtype B and CRF02_AG at (7–7.3%) and 46% in subtype D (Charpentier et al. 2012, Soulié et al. 2013). Importantly, the vast majority of subtype AE viruses harbor the H375 as a natural polymorphism (Figs 7C and 8), the comparatively large histidine residue at this position as compared to serine with other HIV-1 subtypes sterically inhibits temsavir from entering the binding pocket (Zhou et al. 2014, Pancera et al. 2017). In addition, position 375 of HIV-1 gp120 has been shown to have a lower genetic barrier to emergence of S375M/T resistance in CRF02_AG (Fofana et al. 2015). Collectively, these findings emphasize the need for caution when using temsavir in the treatment of non-subtype B strains of HIV-1. Whereas the FDA re-

quired the screening for coreceptor usage with CXCR4 usage preventing use of MVC in cART, temsavir has no treatment restrictions to screen for and prevent treatment in non-subtype B infections. Currently, temsavir is only indicated to treat HIV-1 infection in heavily treatment-experienced adults with multidrug-resistant HIV-1 infection. The FDA approval of course only pertains to the US where over 95% of HIV-1 infections are with subtype B.

The humanized monoclonal antibody ibalizumab, is a first-in-class HIV-1 post attachment inhibitor and became the first monoclonal antibody approved by the FDA for the treatment of HIV-1 in 2018 (Beccari et al. 2019). It is indicated for treatment of PLWH carrying multidrug resistance and failing on cART (FDA 2018). Ibalizumab binds to host cell CD4 receptors to prevent the conformational changes in HIV-1 gp120 necessary for viral fusion (Burkly et al. 1992, Iacob and Iacob 2017). Resistance to ibalizumab is associated with the reduced expression or loss of N-linked glycosylation sites in the gp120 variable loop 5 (V5). These are believed

Table 1. The INSTIs associated resistance mutations frequency and impact on susceptibility in different HIV-1 subtypes.

HIV-1 mutation	Cell culture/patients	Mutation frequency		Reference
		a) Increased frequency	Less frequency	
R263K	Cell culture	B	CRF02_AG, C	(Quashie et al. 2012)
G118R	Cell culture	Subtype C and CRF02_AG	B	(Quashie et al. 2012)
G140S/A	ART naïve/experienced	B	Non-B subtypes ^a	(Doyle et al. 2015)
Q148H/R/K	ART naïve/experienced	B	Non-B subtypes ^a	(Doyle et al. 2015)
E92Q, Q148H/R	ART naïve/experienced	B	Non-B subtypes ^c	(Garrido et al. 2011)
L74M/A/I	ART naïve/experienced	Non-B subtypes ^b	B	(Garrido et al. 2010)
T97A	ART naïve/experienced	Non-B subtypes ^b	B	(Garrido et al. 2010)
V165I	ART naïve/experienced	Non-B subtypes ^b	B	(Garrido et al. 2010)
V201I	ART naïve/experienced	B	Non-B subtypes ^b	(Garrido et al. 2010)
T206S	ART naïve/experienced	B	Non-B subtypes ^b	(Garrido et al. 2010)
K156N	ART naïve/experienced	B	Non-B subtypes ^b	(Garrido et al. 2010)
I203M	ART naïve/experienced	B	Non-B subtypes ^b	(Garrido et al. 2010)
V72I	ART naïve	B	CRF01_AE	(Nguyen, Ruxrungtham and Delaunay 2012)
L10I	ART naïve	B	CRF01_AE	(Nguyen, Ruxrungtham and Delaunay 2012)
A124T	ART naïve	B	CRF01_AE	(Nguyen, Ruxrungtham and Delaunay 2012)
T125K	ART naïve	B	CRF01_AE	(Nguyen, Ruxrungtham and Delaunay 2012)
G140C/S	ART naïve	B, A	CRF01_AE, CRF02_AG, C	(Maiga et al. 2009, Brenner et al. 2011, Nguyen, Ruxrungtham and Delaunay 2012)
V201I	ART naïve	CRF01_AE	B	(Nguyen, Ruxrungtham and Delaunay 2012)
L74M	INSTIs naïve	CRF43_02G	B, C, D and CRF01_AE	(Delicado, Torrecilla and Holguín 2016)
T97A	INSTIs naïve	Group P, J, CRF18_cpx and F2	B, C, D and CRF01_AE	(Delicado, Torrecilla and Holguín 2016)
G163RK	INSTIs naïve	CRF44_BF, CRF46_BF, CRF17_BF, F1, CRF12_BF and CRF29_BF	B, C, D and CRF01_AE	(Delicado, Torrecilla and Holguín 2016)
V151I	ART naïve	B, C	CRF02_AG, A/CRF01_AE	(Maiga et al. 2009, Brenner et al. 2011)
L10I	ART naïve/experienced	C, F, CRF02_AG	D, CRF01_AE	(Garrido et al. 2011)
T124A	ART naïve	D, CRF01_AE, CRF02_AG	B and F	(Garrido et al. 2011)
G193E	ART naïve/experienced	B and F	C, D, CRF01_AE, CRF02_AG	(Garrido et al. 2011)
L101I + T124A	ART naïve	CRF02_AG, and C	B and D	(Garrido et al. 2011)
E92Q/N155H	Cell culture	B (RAL and EVG)	C	(Bar-Magen et al. 2010)
G118R	RAL-experienced	CRF02_AG (RAL and EVG)	B	(Malet et al. 2011)
Q148R, N155H	Cell culture	HIV-O (DTG)	B	(Depatureaux et al. 2015)
G98, E179, C181, R103, T215	Cell culture	HIV-O (EVG)	HIV-M	(Tebit et al. 2016)
G118R	Cell culture	CRF02_AG (RAL)	B and C	(Quashie et al. 2014)
E138K/Q148	Cell culture	CRF02_AG (RAL)	B	(Quashie et al. 2014)

^a A, C, D, F, G, CRF01, CRF02, CRF03, CRF05, CRF35; ^b A, C, D, F, G, H, J, CRF01_AE, CRF02_AG, CRF03, CRF06, CRF09, CRF11, CRF12_BF, CRF13, CRF14_BG, CRF15, CRF18, CRF19; ^c A, C, D, F, G, CRF01_AE, CRF02_AG. In bold, major INSTIs DRMs. In brackets, INSTIs for which there is increased resistance in presence of respective DRMs.

to reduce the steric hinderance effects imposed by the binding of ibalizumab to CD4 (Toma et al. 2011, Pace et al. 2013). Thus, variable susceptibility of ibalizumab may be due to high variability of N-linked glycosylation sites, as observed among different HIV-1 subtypes (Poon et al. 2007). In drug sensitivity assays using a diverse panel of HIV-1 Env pseudoviruses, ibalizumab was shown to be less effective (mean IC₅₀ fold-difference ≥ 5) at neutralizing subtypes D, G, A, and C as compared to subtype B viruses. These studies suggested that differential sensitivity of ibalizumab may be attributed to the differences in variable loop structure and mutation rates observed across group M subtypes that could affect V5 patterns of glycosylation (Gnanakaran et al. 2007, Palmer

and Poon 2019). However, more research is needed to better understand how and if ibalizumab effectively inhibits HIV-1 isolates of different subtypes in tissue culture and in PLWH. Unlike temsavir with IC₅₀ values that can vary > 1000 fold among HIV-1 isolates and subtypes, the variable inhibition with ibalizumab is only ~5-fold to date. However, with NNRTIs, NNRTIs, PIs, and INSTIs, even a 5-fold variability in inhibition of HIV-1 isolates did have profound effects on treatment outcomes. Thus, temsavir and ibalizumab may provide a potential salvage treatment for certain individuals with multidrug resistant HIV-1 but many infected with non-subtype B may be non-responsive to the treatment or may develop rapid drug resistance.

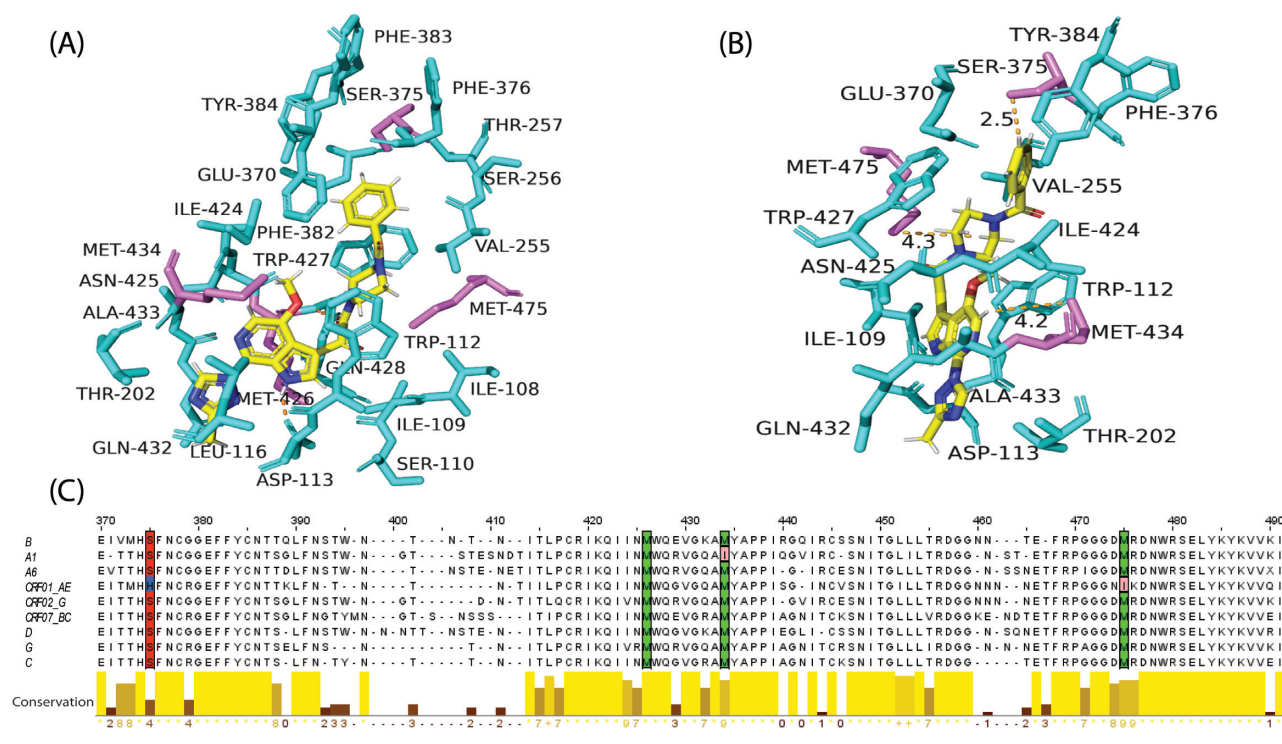


Figure 7. The variability of HIV-1 gp120 amino acid residues at the binding site of BMS- 626529. The HIV-1 gp120 amino acid residues within 5 Å (A) and 3.5 Å (B) of the inhibitor are shown. The carbon atoms of BMS-626529 are depicted in yellow, nitrogen in blue, and oxygen in red. The highly variable amino acid residues S375, M426, M434, and M475 are shown in violet and hydrogen bonds in bright orange. (C) Alignment of HIV-1 gp120 consensus sequences from HIV-1 cART naïve patients, subtype A1 (n = 231), A6 (n = 71), B (n = 6619), C (n = 5516), D (n = 179), G (n = 40), CRF01_AE (n = 1355), CRF02_AG (n = 206), and CRF07_BC (n = 338). The sequences were downloaded from the Los Alamos HIV-1 sequence database and coordinates of HXB2 (K03455) were used to make the alignments.

Maturation inhibitors

Maturation inhibitors (MIs) are an experimental class of HIV-1 antiretrovirals that target processing steps late in the replication cycle, wherein non-infectious particles undergo the structural changes necessary to become infectious virions. These inhibitors do not bind directly to the HIV-1 protease and thus function via mechanisms distinct from PIs. To date, no MIs have been granted FDA approval for the treatment of HIV-1, but several compounds have previously, and are currently, being assessed in clinical trials. Two distinct groups of MIs have been studied extensively and show potential therapeutic promise: 1) compounds that prevent processing of the mature capsid (CA) precursor from spacer protein-1 and 2) capsid-binding compounds that interfere with capsid assembly. Via distinct mechanisms, these MIs prevent the formation of the mature capsid core and thus abrogate HIV-1 infectivity. Since the use of these compounds have been limited to testing involving HIV-1 subtype B isolates in vitro and those infected with subtype B in clinical trials, little is known about the efficacy of these compounds at inhibiting non-subtype B HIV-1. Here, we emphasize the continued need to screen non-subtype B HIV-1 in the development and testing of MIs.

The first-in-class MI bevirimat (BVM), formerly (PA-457) [3-O-(3',3'-dimethylsuccinyl) betulonic acid], blocks the processing of CA-SP1 to CA. BVM prevents protease-mediated cleavage at CA-SP1 in immature HIV-1 particles by binding directly to the CA-SP1 cleavage site on assembled Gag hexamers and stabilizing the bundle (Sakalian et al. 2006, Nguyen et al. 2011). Stabilization of the Gag bundle structure prevents protease from accessing the CA-SP1 junction buried within the oligomer (Schur et al. 2016). Mu-

tations that confer resistance to BVM flank the CA-SP1 junction (Adamson et al. 2006) but are highly conserved residues across the major group M subtypes (Yebra and Holguin 2008). BVM was effective at reducing viral load in individuals infected with subtype B consensus SP1. However, in a subset of PLWH, natural polymorphisms of HIV-1 Gag positions Q369, V370, and T371 conferred intrinsic resistance in approximately 50% of PLWH who did not respond to BVM in phase 2b clinical trials (Lalezari et al. 2008). The frequency of these natural polymorphisms of Gag positions Q369, V370, and T371 appear to be higher in non-B HIV subtypes than subtype B viruses (93.1% vs. 39.3%; $P < 0.001$) (Seclen et al. 2010). Phenotypic analyses of patient-derived viruses carrying these mutations showed > 100-fold resistance to bevirimat (Van Baelen et al. 2009). BVM was even less effective at inhibiting non-subtype B HIV-1 in vitro and in small-scale clinical trials due to even higher variability at the SP1 sites (Van Baelen et al. 2009, Lu et al. 2011). This intrinsic resistance was observed in > 90% of non-subtype B but the intrinsic resistance with ~30% subtype B HIV-1 halted development of BVM as an HIV-1 therapeutic.

Second generation maturation inhibitors, analogs of BVM, retain potency against the CA-SP1 polymorphisms resistant to BVM (Dang et al. 2013, Liu et al. 2016, Urano et al. 2016). Mutations that confer resistance to these second-generation MIs were selected for in subtype B and C HIV-1 isolates by Urano et al (Urano et al. 2018). This work uncovered novel mutations in CA (CA-V230M) and in SP1 (SP1-A1V, S5N, and G10R) that were observed exclusively in the subtype C HIV-1 selection experiments and confer modest to high-level resistance to the second-generation MIs tested. The results of phase I and II clinical trials for the two known GlaxoSmithKline second-generation MIs (GSK2838232 and

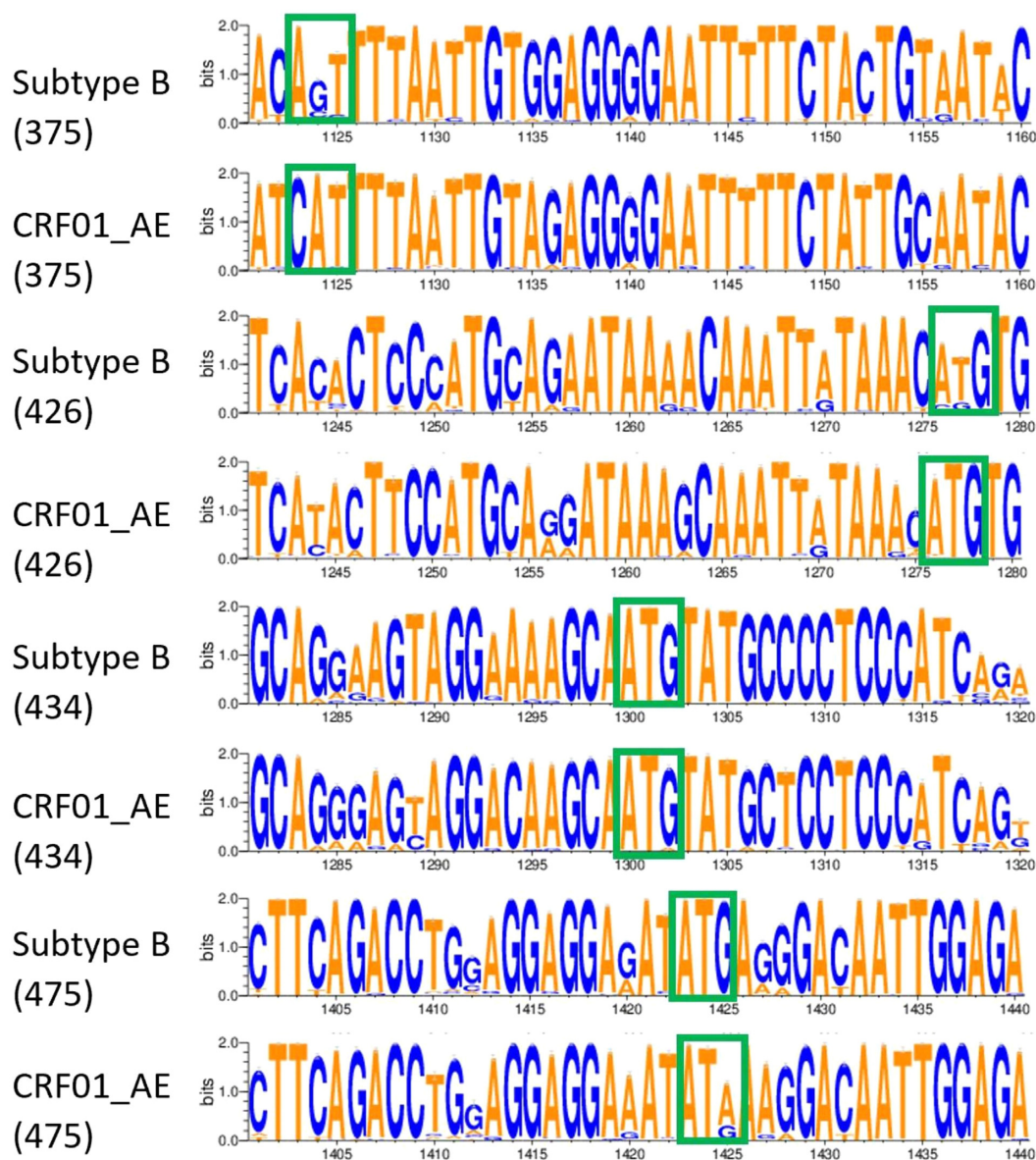


Figure 8. The variability of HIV-1 gp120 consensus sequences between HIV-1 subtype B and CRF01_AE. The codon sequences coding for temsavir active site amino acids 375, 426, 434, and 475 are boxed in green. The subtype B ($n = 6619$) and CRF01_AE ($n = 1355$) gp120 sequences from HIV cART naïve patients were downloaded from the Los Alamos HIV sequence database and sequence position coordinates of the HXB2 sequence (K03455) were used to make the alignments.

GSK3640254) have shown short-term antiviral activity and tolerability, providing a positive proof of concept for further long-term studies (Edwin DeJesus et al. 2019, Mark Mascolini 2019). Again, little is known about their efficacy for the > 90% of HIV-1 which are not subtype B. The work by Urano and colleagues emphasizes the importance of testing these and other novel MIs in non-subtype B HIV-1 prior to wide-scale approval (Urano et al. 2018).

Disruption of capsid assembly and/or disassembly has anti-HIV-1 therapeutic potential, and several CA inhibitors have been developed and studied extensively (Tang et al. 2003, Sticht et al. 2005, Blair et al. 2010, Lemke et al. 2012). The Gilead Sciences compound, GS-CA1 is a small molecule that targets HIV-1 CA and interferes with capsid-mediated nuclear import of viral DNA preventing later assembly of the virus core. GS-CA1 is a long-acting agent in both mouse models and human studies with potential for treatment of HIV-1 infection (Tse et al. 2017, Yant et al. 2019).

Phenotypic analyses have shown that CA mutations L56I, M66I, Q67H, N74D, and A105E reduce susceptibility to GS-CA1, but with reduced replication fitness (Tse et al. 2017). Fortunately, these mutations have not been detected as naturally occurring polymorphisms in subtype B and non-B subtype HIV-1 in PLWH naïve to treatment (Perrier et al. 2017). Naturally occurring substitution F169, observed in HIV-1 subtype C viruses, is in close proximity to the GS-CA1 binding site, which may weaken interactions with the inhibitor (Singh et al. 2019).

Lenacapavir (formerly known as GS-6207) is another compound being developed by Gilead Sciences in favor over GS-CA1, with similar mechanism of action, potency, and metabolic activity (Bester et al. 2020, Link et al. 2020). Lenacapavir binds directly to the assembled CA hexamer to inhibit multiple sequential steps during virus ingress and during late stages of the replication cycle. Lenacapavir is first-in-class of capsid inhibitors to enter clini-

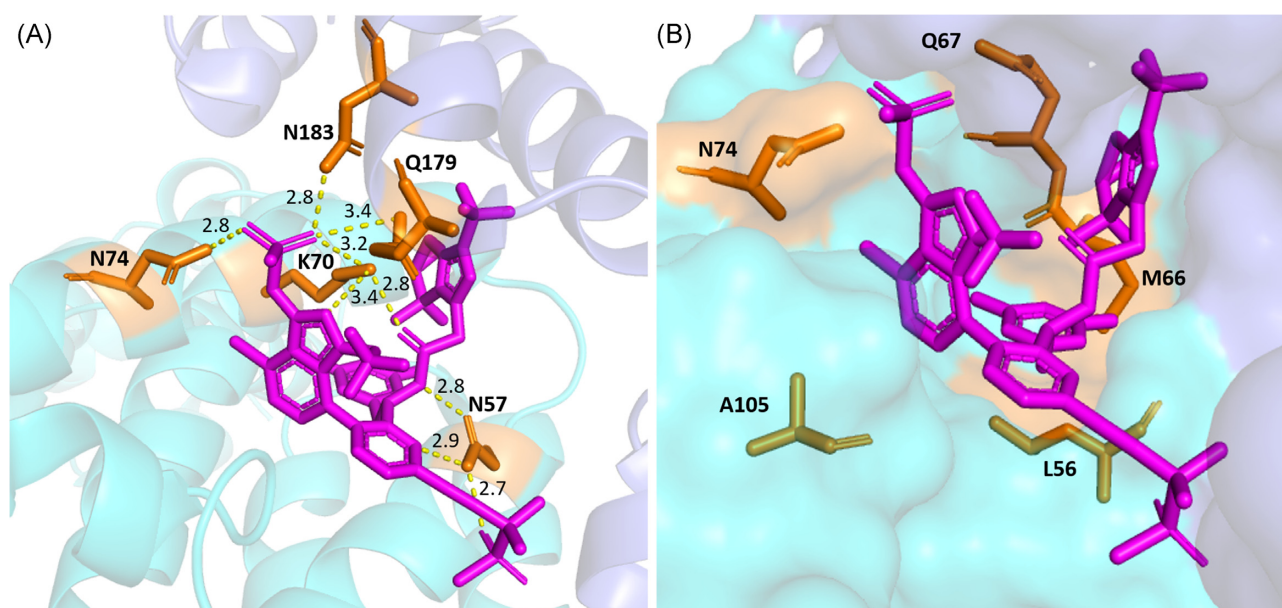


Figure 9. HIV-1 CA and GS-6207 complex. X-ray crystal structure of GS-6207-CA (PDB ID 6V2F) (Link et al. 2020) shows novel capsid inhibitor GS-6207 binding between NTD of one CA monomer (cyan) and CTD of a neighboring monomer (dark blue) to interfere with assembly and disassembly of HIV-1, leading to production of non-infectious virions. **(A)** GS-6207, depicted in magenta, interacts with residues (orange) N57, K70, N74 of the CA-NTD monomer, as well as Q179 and N183 of the adjacent CA-CTD monomer. **(B)** The residues associated with resistance against GS-6207 are depicted in orange (L56, Q67, M66, N74, A105) and are localised at the active site of GS-6207.

cal trials and is being investigated as a long-acting agent in cART naïve and experienced PLWH with multi-drug resistance (CALIBRATE and CAPELLA) studies (ClinicalTrials.gov 2019a,b). CALIBRATE is an open-label phase II trial of 182 ART naïve participants with HIV-1 RNA ≥ 200 copies/ml, CD4 count ≥ 400 cells/ μ l and no hepatitis B or C virus. At week 54, 87% participants (average of 3 test groups) and 92% participants in the control group had achieved HIV RNA < 50 copies/ml (Gupta et al. 2022). There was one case of CA emerging resistance (Q67H/K70R) at week 10. In the phase 2/3 CAPELLA trial assessing lenacapavir in 72 heavily cART-experienced patients with multidrug resistance, 36 participants were randomly assigned to add either oral lenacapavir or a placebo to their failing regimen for 14 days, thereafter lenacapavir every after 6 months plus optimised background regimen based on genotypic results (cohort 1). In cohort 2, 36 participants were given lenacapavir plus an optimised background regimen from the outset. At the end of initial 14-day period (primary end point), 88% of patients in lenacapavir group had HIV-1 RNA drop ≥ 0.5 log₁₀ copies/mL compared to 17% in the placebo group ($P < 0.0001$), and by week 52, 83% of participants had HIV RNA < 50 copies/ml in cohort 1 (Ogbuagu et al. 2022). At 26 weeks, lenacapavir resistance mutations, M66I, M66M/I, N74D, A105T, and K70K/S were detected in 11% (4/36) participants all of which were associated with effective lenacapavir monotherapy.

An X-ray crystal structure of lenacapavir bound to HIV-1 CA hexamer shows hydrogen bond interactions with the inhibitor through residues N57, K70, N74, Q179, and N183 in CA (Link et al. 2020) (Fig. 9A). The residues interacting with lenacapavir are mostly conserved across HIV-1 subtypes A1, B, C, D, CRF07_BC, and CRF01_AE isolates. However, as with resistance to PIs, amino acid residues with direct contact to the inhibitor rarely mutate to confer resistance because these mutations result in dysfunctional or dead virus.

Lenacapavir has also shown antiviral activity against susceptible and antiretroviral resistant variants across different HIV-1

subtypes. This observation, coupled with its prolonged plasma half-life as an injectable and strong safety profiles, support its use as a long-acting cART agent to treat or prevent HIV-1 (Link et al. 2020). Selection experiments with subtype B HIV-1 show reduced susceptibility (6–3200-fold) to lenacapavir in presence of capsid mutations, M66I, Q67H, and N74D as major/primary resistance mutations, and L56I, K79N, Q67H/N74S, and Q67H/T107N as possible secondary mutations (Link et al. 2020). These mutations occur in the CA binding site for lenacapavir, and all show reduced viral fitness except Q67H. The M66I mutation has by far the greatest impact on susceptibility to lenacapavir, reducing lenacapavir potency by over 1000-fold (Bester et al. 2020), and is preferentially selected in highly treatment experienced PLWH during periods of effective lenacapavir monotherapy (Margot et al. 2021). These resistance-associated residues (Fig. 9B) are however conserved across HIV-1 group M viruses (Li et al. 2013a, Perrier et al. 2017, Yant et al. 2019). Moreover, deep sequencing of CA from both subtype B and non-B infected, treatment-naïve PLWH revealed very low ($< 1\%$) prevalence of pre-existing HIV-1 variants with lenacapavir resistance substitutions, and significantly, none with the M66I substitution (Bester et al. 2020). Based on recent deposits of CA sequences from LMICs in the Los Alamos HIV database, our analysis of 18 379 sequences from the database shows nearly 100% conservation in subtype A. However, there is a very low prevalence of ‘resistant’ sequence as natural polymorphism H67 (2.16%) and D74 (0.59%) in HIV subtype D, as well as H67 (0.2%) in subtype CRF01_AE viruses from ART naïve PLWH (Table 2). Nonetheless, these findings support the clinical development of lenacapavir as a potential inhibitor of most HIV-1, regardless of subtype. However, further research of screening non-subtype B HIV-1 in phenotypic susceptibility assays with lenacapavir is warranted to confirm relatively invariable inhibition profile (as observed with most NNRTIs, NRTIs, and PIs) and/or dissect implications of active site substitutions on lenacapavir efficacy (especially N183 on lenacapavir efficacy in the context of non-

Table 2. Amino acid conservation of HIV CA residues at GS-6207 active site and those associated with GS-6207 resistance.

Subtype	No	L56	M66	% Q67	N74	A105	N57	K70	N183
A	7	100	100	100	100	100	100	100	100
A2	2	100	100	100	100	100	100	100	98
A1	1565	99.2	99.8	99.8	99.9	99.9	99.7	97.8	32.9
B	6426	99.3	99.9	99.3	99.7	99.5	99.1	99.5	97.3
C	6606	98.5	99.9	99.6	99.8	99.7	99.5	99.5	95.9
D	508	97.8	99.8	95.8	95.6	99.8	98.6	96.6	87.2
G	40	100	100	100	100	100	100	100	57.5
F1	239	100	100	100	100	96.6	100	100	68.6
F2	1	100	100	100	100	100	100	100	100
H	1	100	100	100	100	100	100	100	100
N	3	100	100	100	100	100	100	100	100
O	2	100	100	100	100	100	100	100	100
P	4	100	0	100	100	100	100	100	100
U	4	100	100	100	100	100	100	100	100
CRF73_BG	1	100	100	100	100	100	100	100	100
CRF47_BF	2	100	100	100	100	100	100	100	100
CRF41CD	3	100	100	100	100	100	100	100	100
CRF37_cpx	1	100	100	100	100	100	100	100	100
CRF36_cpx	4	100	100	100	100	100	100	100	50
CRF35_AD	9	100	100	100	100	100	100	100	100
CRF29_BF	5	100	100	100	100	100	100	100	25
CRF28_BF	3	100	100	100	100	100	100	100	100
CRF25_cpx	2	100	100	100	100	100	100	100	50
CRF21_A2D	1	100	100	100	100	100	100	100	100
CRF16_A2D	3	100	100	100	100	100	100	100	100
CRF14_BG	3	100	100	100	100	100	100	100	100
CRF12_A2D	21	100	100	100	100	100	100	100	40
CRF08_BC	47	100	100	100	100	100	100	100	100
CRF06_cpx	14	100	100	100	100	100	100	100	100
CRF03_AB	1	100	100	100	100	100	100	100	100
CRF02_AG	120	100	100	100	100	100	100	100	87.5
CRF07_BC	655	100	100	100	99.8	99.8	99.8	100	99.6
CRF01_AE	2076	100	99.9	99.8	100	99.6	99.6	99.8	94.5

HIV-1 CA sequences were retrieved from 18379 ART naïve patients infected with different HIV subtypes and recombinant forms (<https://www.hiv.lanl.gov/components/sequence/HIV/search/search.html>). Green depicts Aas conservation of residues associated with GS-6207 resistance and orange depicts Aas conservation of residues that interact with GS-6207.

subtype B HIV-1). Also, evidence of poor replicative fitness of drug resistance mutations in one HIV-1 subtype does not always related to same in another subtype. Case in point, HIV-1 subtype C integrase proteins show low cost on enzymatic fitness due to G118R mutation compared to subtype B or recombinant CRF02_AG proteins (Quashie et al. 2014).

Small molecules targeting HIV-1 accessory proteins

Over the years, the role of HIV-1 accessory proteins in regulating HIV-1 replication and infectivity has been studied with increasing clarity, assisting in identifying potential drug targets. Potential candidates currently under study include the small molecules BIT225 (N-[5-(1-methyl-1H-pyrazol-4-yl)-naphthalene-2-carbonyl]-guanidine) and ABX464, which target HIV-1 Vpu and Rev accessory proteins, respectively. BIT225 specifically inhibits the viroporin activity of HIV-1 vpu. BIT225 inhibits both Hepatitis C Virus and HIV-1 by blocking viral particle release from the cell membrane (Luscombe et al. 2010, Kuhl et al. 2011) and inhibits release of HIV-1 from infected macrophages. A phase1a/2b trial has shown its ability to cross the blood-brain barrier, which may reduce the formation of viral reservoirs (Khoury et al. 2010, Wilkinson et al. 2016). Unfortunately, in a phase 2 clinical trial of 36 cART

naïve PLWH, there was similar decline of HIV-1 RNA levels in both BIT225 and placebo groups, but the BIT225 group did have significant increases in CD8+, CD4+, and NK cells compared to placebo (Luscombe et al. 2020). A further clinical trial is planned to confirm these results and will include both ART naïve and experienced individuals (Luscombe et al. 2020).

ABX464, initially developed for its anti-inflammatory effects in inflammatory bowel disease, is a first-in-class agent that targets the cap-binding complex required for proper cellular and viral pre-mRNA splicing and ribosomal binding (Campos et al. 2015). ABX464 has been shown to inhibit HIV-1 by enhancing HIV-1 mRNA splicing; it abrogates the ability of HIV-1 Rev to rescue singly spliced and unspliced mRNA from nucleases through the CRM1-mediated pathway (Vautrin et al. 2019). In effect, ABX464 decreases the production of the HIV-1 Gag and Env proteins in favour of the accessory proteins. In phase 2 clinical trials with HIV-1 infected cART naïve PLWH, ABX464 was safe and well tolerated with reduction in HIV-1 RNA demonstrated at higher doses of inhibitor (Steens et al. 2017). Preliminary findings in both human clinical trials and humanised mouse models using ABX464 also suggest a reduction in latent HIV-1 reservoirs, supporting the potential of ABX464 use as a functional cure therapeutic (Campos et al. 2015, Rutsaert et al. 2019). Although preliminary results appeared promising, ABX464 development appears stalled

and whether ABX464 has subtype-specific anti-HIV-1 activity requires further investigation.

Summary

Much remains unknown regarding the broad effects of HIV-1 subtype on the ultimate treatment responses to cART and emergence of drug resistance in PLWH, mostly in resource limited settings. Collectively, the literature demonstrates a variable efficacy of approved treatment regimens across diverse HIV-1 strains through a very limited set of trials or post-drug approval studies. No antiretroviral drugs to date were designed for inhibition on non-subtype B HIV-1 (~33 million infections) but rather developed as inhibitors of subtype B HIV-1 (3–4 million) almost exclusively living in high income countries. Clearly, lack of adherence to cART is the major factor responsible for treatment failure and drug resistance. Many had predicted reduced cART adherence in LMICs versus HICs would be due to perceived behavioural differences but in fact the opposite is true if the same cART is readily available. Reduced adherence in LMICs versus HICs is not related to willingness to take the medication but rather, is due to socioeconomic factors leading to poor clinic and drug access as well as intermittent reductions in drug supply. Nonetheless, there is also the factor that is rarely addressed in the rollout of ARVs in LMICs. Most ARVs go through minimal and ineffective testing in LMICs to determine the efficacy on diverse HIV-1 subtypes, among diverse human populations, and in the background of different and more endemic secondary infections. Fortunately, most ARVs to date have reasonable potency against non-subtype B HIV-1. However, the evolution to treatment failure and drug resistance has differed between subtype B and non-subtype B HIV-1. In most instances, resistance in non-subtype B HIV-1 is faster to emerge (possibly related to poorer adherence) and more difficult to discern based on the lack of resistance testing. There is also the added complication of using drug mutational patterns derived from HIV-1 subtype B and applied to non-subtype B sequences that can differ 5–15% in amino acid diversity within the targeted HIV-1 proteins/enzymes.

Regardless of differential resistance patterns, income inequality and reduced access to the best cART has driven a rapid increase in drug resistance across LMICs. As such, more second, third- and fourth-line treatment regimens are necessary in these LMICs. While d4T was the mainstay for first line cART for all LMICs until 2010, and NVP (or EFV) + AZT (or TDF) + 3TC (or FTC) until 2017/18, the rest of the world were employing better tolerated cART involving INSTIs with higher genetic barriers for resistance. With first line cART in LMICs (pre-2017), approximately 10% per year were failing treatment and harbored NNRTI-resistant virus. Thus, there is a growing HIV-infected population in the world with drug resistance that can only be treated with new inhibitors targeting entry, attachment, and assembly. Maraviroc, ibalizumab, and temasvir are FDA approved for highly ARV experienced individuals with multi-ARV resistance. However, unlike the former ARVs in the NNRTI, NRTI, PI, and INSTI classes, it is unlikely that these new inhibitors will be effective as salvage for a large proportion (even majority) infected with non-subtype B HIV-1 (even if these drugs are provided to LMICs). Only the most recent assembly inhibitors (e.g. lenacapavir) show strong cross-subtype inhibition and may be a promising new ARV for the future of cART in LMICs.

Clearly, a more dominant subtype, like HIV-1 subtype C (>15 million infections) could be used as the prototype HIV-1 isolate for screening and development of new antiretrovirals. However, the potential of the high revenue based on drug screening against

subtype B HIV-1 isolates provides a stronger financial incentive than basing drug development on the more common subtypes A, C, and even D HIV-1. With the NNRTI, NRTI, PI and INSTI drugs, a less than perfect arrangement with the pharmaceutical industry not fighting production of generic drugs for LMICs has helped to provide treatment for nearly 20 million around the world. Our best hope is that development of new ARVs will place more emphasis on broad inhibition of all HIV-1 subtypes, carry out clinical trials of these inhibitors in diverse cohorts, and to screen for resistance in all individuals.

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